

CS 2614: Computer Organization

Lab 2

Spring 2025

K-Maps with Don't Cares

Student ID	Name



***The* UNIVERSITY of OKLAHOMA**

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Pre-Lab Exercise

1. Using a K-Map, simplify the Boolean function $F(A, B, C, D) = \sum (0,1,2,3,7,8,10)$.

2. Using a K-map and the *don't-care conditions* d , simplify the Boolean function F in *Sum-Of-Products* form.

$$F(A, B, C, D) = \sum (0,1,2,3,7,8,10)$$

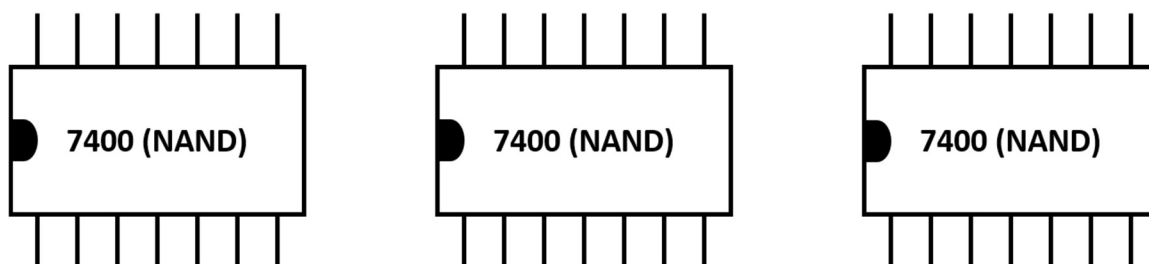
$$d(A, B, C, D) = \sum (5,6,11,15)$$

5. Draw the logic diagram for **problem 2** using only NAND gates. You must not use any OR, NOT, or AND gates. All used NAND gates should have only 2 input pins.

Remember to upload this section to the Canvas!

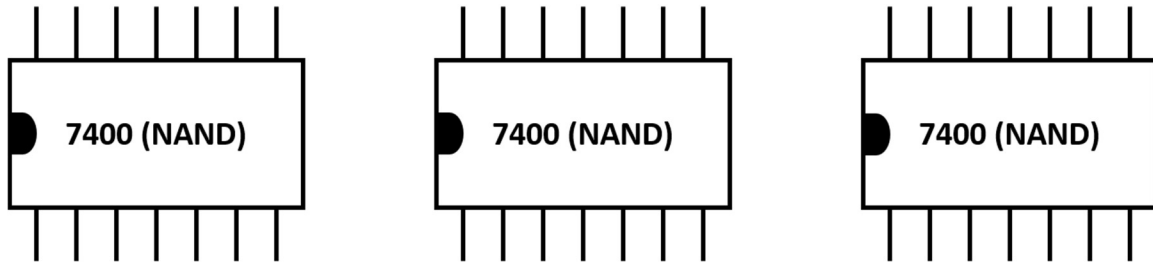
In-Lab Exercise

6. Draw the circuit from Problem 4 using **NAND gates** (label each input/output, some chips may not be needed).



7. Implement the circuit from Problem 6 on the breadboard.

8. Draw the circuit from Problem 5 using **NAND gates** only on the chips below (label each input/output, some chips may not be needed).



9. Implement the circuit from Problem 8 on the breadboard.